INGENIERIE SYSTEME

2026-2027 - Introduction

ENSTA | INSTITUT POLYTECHNIQUE DE PARIS



Cours-projet condensé (60h réparties sur 3 semaines)

- 4 groupes de 6 étudiants
- Grande diversité de tâches

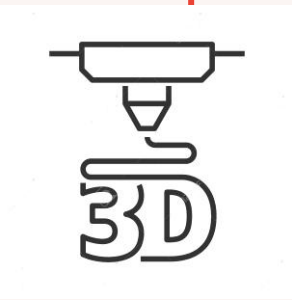
} Focus sur le travail d'équipe

Simulation dynamique

Cours-exercices
Projet

Conception Mécanique

Cours
Projet



Ingénierie système

Cours-exercices
RDV clients

Emploi du temps

Semaine 42	lundi 12 oct.	mardi 13 oct.	mercredi 14 oct.	jeudi 15 oct.	vendredi 16 oct.
ROB					
Semaine 43	lundi 19 oct.	mardi 20 oct.	mercredi 21 oct.	jeudi 22 oct.	vendredi 23 oct.
ROB					
Semaine 44	lundi 26 oct.	mardi 27 oct.	mercredi 28 oct.	jeudi 29 oct.	vendredi 30 oct.
ROB					
Semaine 45	lundi 2 nov.	mardi 3 nov.	mercredi 4 nov.	jeudi 5 nov.	vendredi 6 nov.
ROB					
Semaine 45	lundi 9 nov.	mardi 10 nov.	mercredi 11 nov.	jeudi 12 nov.	vendredi 13 nov.
ROB					
Semaine 45	lundi 16 nov.	mardi 17 nov.	mercredi 18 nov.	jeudi 19 nov.	vendredi 20 nov.
ROB					

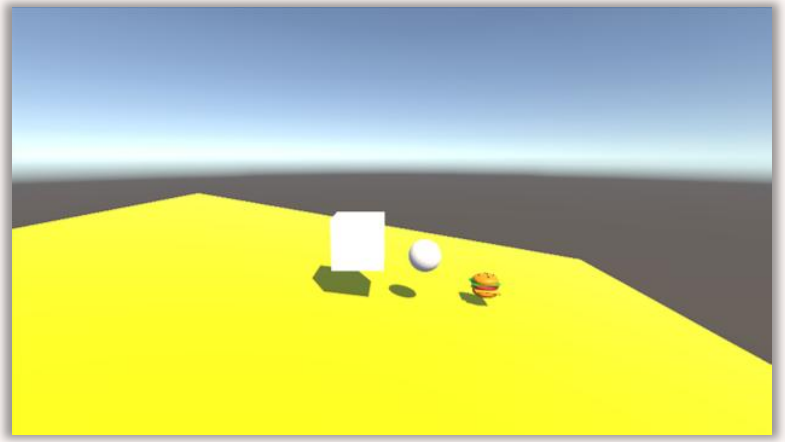
Apprentissage de Unity & ROS 2

Projet Simulation + Créneaux Méca

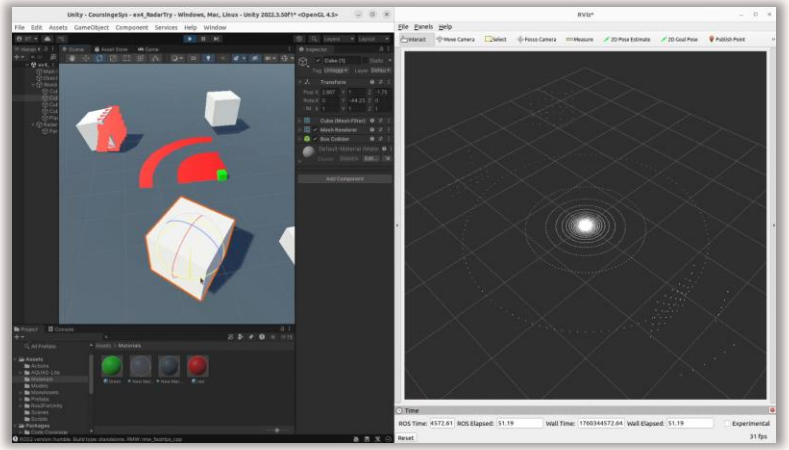
RDV clients

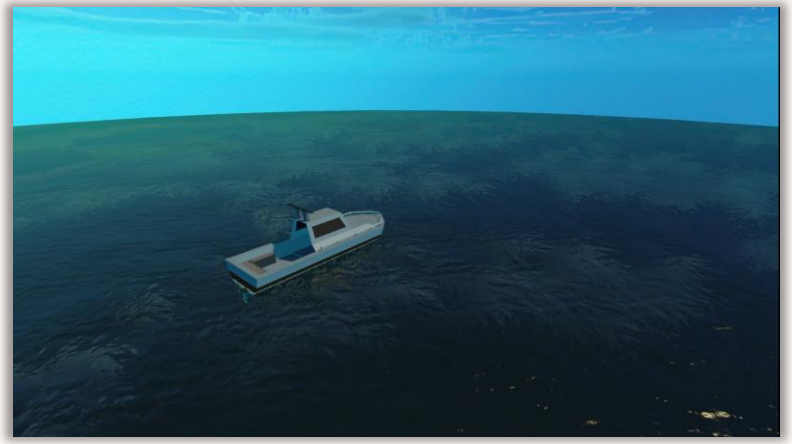
Présentation Orale Finale

Simulation réalistes à réaliser durant ce cours









Dynamical Simulation

Mathematical Representation of a Dynamic System

A dynamic system can be represented by a set of ordinary differential equations (ODEs):

$$\dot{x}(t) = f(x(t), u(t), w(t), t)$$

$$y(t) = h(x(t), u(t), w(t), t)$$

where:

— $x(t) \in \mathbb{R}^n$ is the state vector (e.g., position, velocity, orientation, etc.),

— $u(t) \in \mathbb{R}^m$ is the control vector,

— $w(t)$ represents external disturbances or unmodeled dynamics,

— $y(t) \in \mathbb{R}^n$ is the output vector (measured quantities, sensor outputs),

— f and h are non linear functions representing respectively, the system dynamics and the measurement model.

The function f governs the evolution of the state over time given the current state, input, and possible disturbances.

Simulator

A simulator is a digital tool designed to reproduce the behavior of a physical system from either a mathematical model or a physical model.

Its purpose is to study, test, and visualize how a robot behaves without the need to manipulate the real system.

In mobile robotics, a simulator allows you to:

— Predict the trajectory and forces acting on the robot as a function of its control inputs,

— Validate a control or navigation algorithm,

— Or simply visualize the robot's motion in a 3D environment.

This simulator can be seen as a function $S : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ computing the next state x_{k+1} according to the actual state x_k and its derivative $\dot{x}_k$.

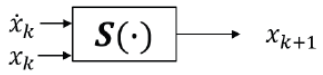


Figure 1: Simulator diagram

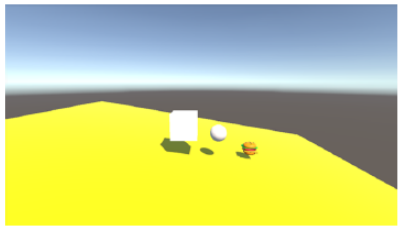


Figure 3: Summary illustration of the proposed approach

In a 2-D fixed frame $\mathcal{R}$. The following notations will be used throughout this paper:

— $X(t) = [x(t), y(t)]^T$: vehicle's position in $\mathcal{R}$;

— ψ: vehicle's heading;

— v: vehicle's speed;

— R: vehicle's minimum turning radius;

$$\begin{cases} \dot{X}(t) & = \begin{pmatrix} u_1 \cos(\psi(t)) \\ u_1 \sin(\psi(t)) \end{pmatrix} \\ \dot{\psi}(t) & = u_2(t) \end{cases} \quad (1)$$

where $u_1(t) \in [0, v_{max}]$ is the forward speed control input and $u_2(t) \in \left[-\frac{R}{v_{max}}, \frac{R}{v_{max}}\right]$ is the turning speed control input.

In this exercise, we won't use a Rigidbody but only a C# script modifying the car state at every step. The state of an object can easily be changed by modifying its transform.

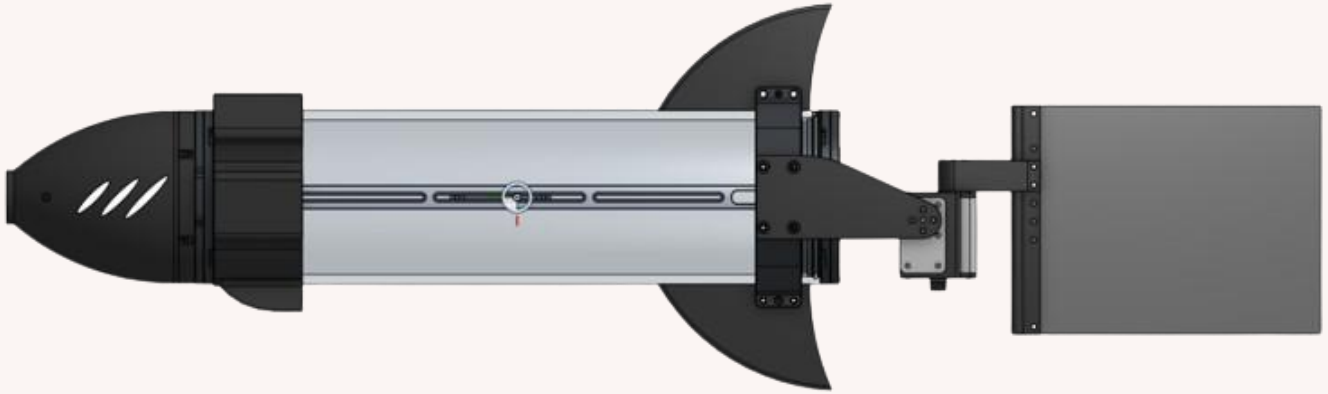
— Place a plane and a car model in your Unity scene, add it a C# script;

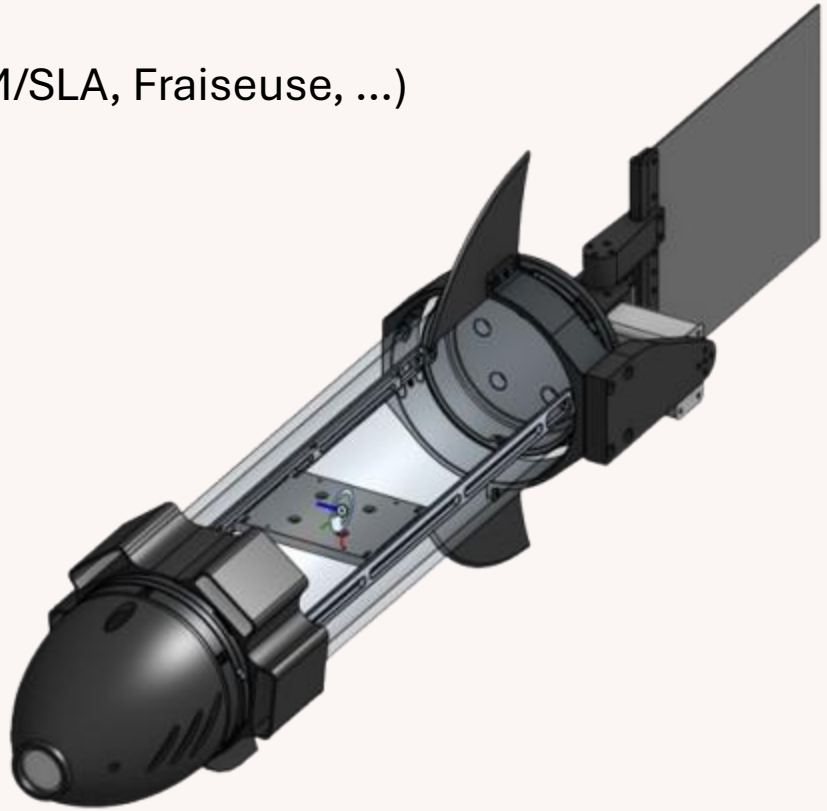
— Add the Dubins' car model in your C# script. Do not forget Unity object's frame is not standard and, particularly, is indirect;

— Use Unity input system to be able to control u_1 and u_2 at any time. Recently, Unity developped a new input system which is both more complex and complete. It is recommended to learn how to use it.

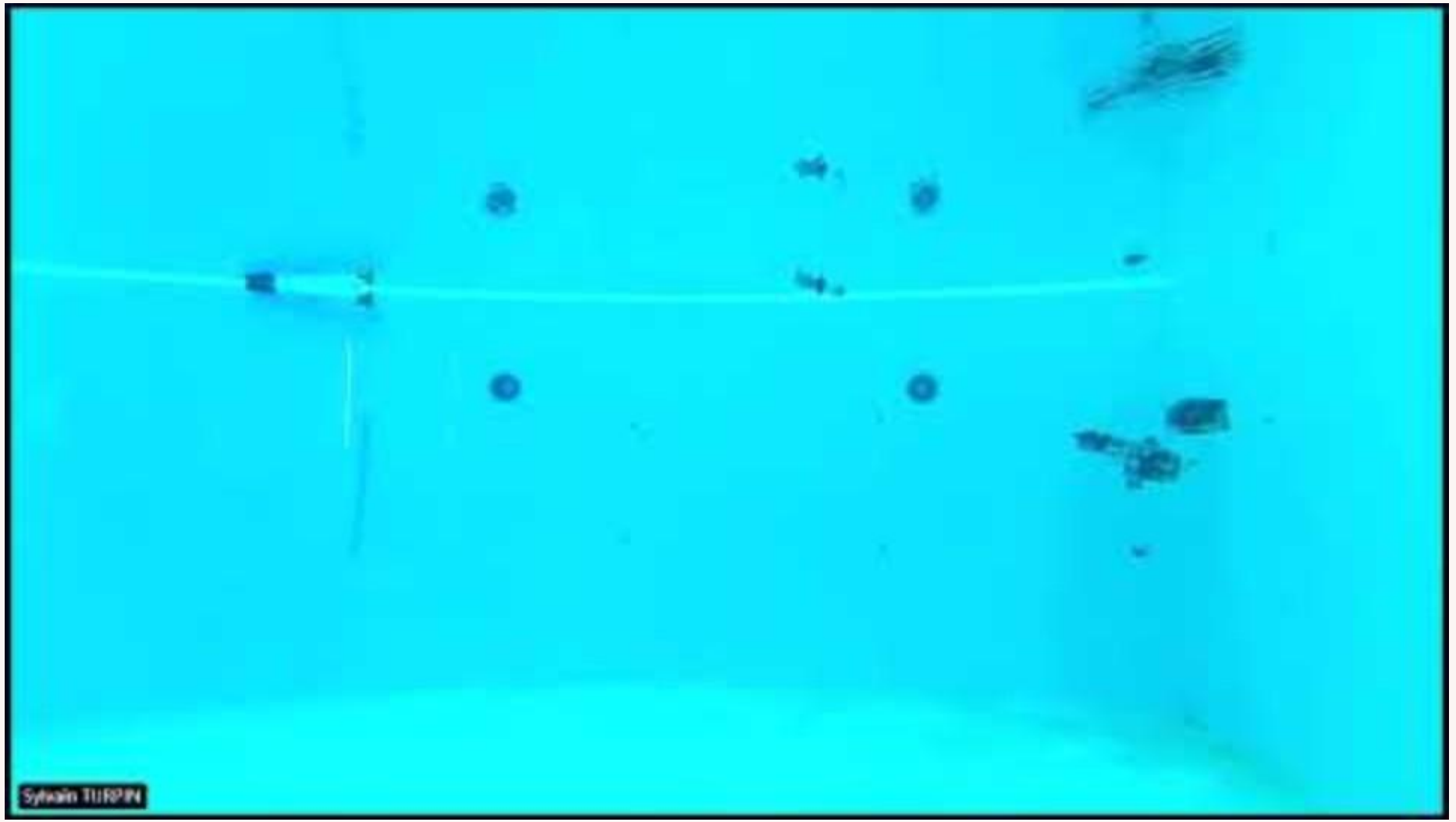
Remark : In this exercise, we implement a kinematic C# script to move an object by its transform. However, this way of doing it do not allow the user to collide with other 3D object of the scene without creating all the codes manually. Next exercise propose to use Unity dynamics model.

- Fabrication d'un robot par groupe de 5 étudiants
- CAD de démarrage fournie
- Utilisation de Machine-Outils numériques (Imprimantes 3D FDM/SLA, Fraiseuse, ...)





https://www.youtube.com/watch?v=F_cFXFzy57Q



https://www.youtube.com/watch?v=F_cFXFzy57Q

Notations

Code	Rubrique	Critère	Poids critères dans rubrique	Poids rubrique dans note finale (/)	Min (/20)	Max (/20)		Groupe 1 (/20)	Groupe 2 (/20)	Groupe 3 (/20)	Groupe 4 (/20)	Groupe 5 (/20)	Groupe 6 (20)
1	Valeur Technique - SIMULATEUR	Unity	1	1/3	0	20							
2	Valeur Technique - SIMULATEUR	Unity & ROS2	1		0	20							
3	Valeur Technique - SIMULATEUR	Respect du cahier des charges	1		0	20							
4	Valeur Technique - SIMULATEUR	Démonstrateur	1		0	20							
5	Valeur Technique - SIMULATEUR	Bonus = Fonctionnalités en plus	Ajout [0,1] à note finale		0	1							
					0	21	TOTAL SIMULATEUR	0	0	0	0	0	0
5	Valeur Technique - ROBOT	Fonctionnalités matérielles ?	1	1/3	0	20							
6	Valeur Technique - ROBOT	Fonctionnalités logicielles ?	1		0	20							
7	Valeur Technique - ROBOT	Respect du cahier des charges	1		0	20							
8	Valeur Technique - ROBOT	Vidéo	1		0	20							
8	Valeur Technique - ROBOT	Bonus = Fonctionnalités en plus	Ajout [0,1] à note finale		0	1							
					0	21	TOTAL ROBOT	0	0	0	0	0	0
9	Ingéniérie Système	Outil collaboratif GitLab - structure & organisation	1	1/3	0	20							
10	Ingéniérie Système	Travail en équipe - répartition des tâches & collaborations	1		0	20							
11	Ingéniérie Système	Entretiens Clients (Rappels, supports, tâches réalisées et a réaliser, restitution à l'équipe)	1		0	20							
12	Ingéniérie Système	Oral final - Présentation Qualité & Structure	1		0	20							
13	Ingéniérie Système	Oral Final - Prise de recul sur le travail réalisé	1		0	20							
					0	20	TOTAL IS	0	0	0	0	0	0

Objectifs de la première séance

- Création de 5 équipes
- Installation et première utilisation de Unity et ROS2
- Création d’un GitHub par équipe, ajouter les encadrants en tant qu’observateur

Thank you for your attention

Do you have any question?

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